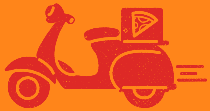
PIZZA SALES ANALYSIS PROJECT

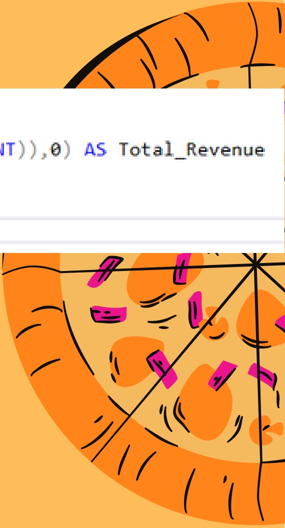


Project Objective

ANALYZE PIZZA SALES DATA TO IDENTIFY REVENUE DRIVERS, CUSTOMER ORDERING PATTERNS, AND OPPORTUNITIES TO INCREASE SALES AND REDUCE OPERATIONAL INEFFICIENCIES.

1. WHAT IS THE TOTAL REVENUE GENERATED BY THE PIZZA SHOP?

```sql
SELECT
    COALESCE(SUM(CAST(p.price AS DECIMAL(10,2)) * CAST(od.quantity AS INT)),0) AS Total_Revenue
FROM order_details AS od
JOIN pizzas AS p
ON od.pizza_id = p.pizza_id;
```



RESULT!

	Total_Revenue
1	1635720.10

2. WHICH PIZZA CATEGORY CONTRIBUTES THE MOST REVENUE?

```sql
SELECT TOP 1
      pt.category,
      COALESCE(SUM(CAST(p.price AS DECIMAL(10,2)) * CAST(od.quantity AS INT)), 0) AS Highest_Rev_By_category
FROM pizza_types AS pt
INNER JOIN pizzas AS p
ON pt.pizza_type_id = p.pizza_type_id
INNER JOIN order_details AS od
ON od.pizza_id = p.pizza_id
GROUP BY pt.category
ORDER BY Highest_Rev_By_category DESC;
```

RESULT!

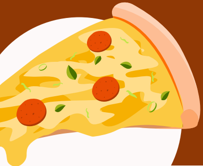
	category	Highest_Rev_By_category
1	Supreme	416394.00

3. WHAT ARE THE TOP 5 BEST-SELLING PIZZAS BY REVENUE?

```sql
SELECT TOP 5
      pt.name,
      COALESCE(SUM(CAST(p.price AS DECIMAL(10,2)) * CAST(od.quantity AS INT)), 0) AS Best_Selling_By_Rev
FROM pizza_types AS pt
INNER JOIN pizzas AS p
ON pt.pizza_type_id = p.pizza_type_id
INNER JOIN order_details AS od
ON od.pizza_id = p.pizza_id
GROUP BY pt.name
ORDER BY Best_Selling_By_Rev DESC;
```

RESULT!

	name	Best_Selling_By_Rev
1	The Thai Chicken Pizza	86868.50
2	The Barbecue Chicken Pizza	85536.00
3	The California Chicken Pizza	82819.00
4	The Classic Deluxe Pizza	76361.00
5	The Spicy Italian Pizza	69662.50



4. WHICH PIZZAS GENERATE THE LEAST REVENUE (SLOW MOVERS)?

```sql
SELECT TOP 3
       pt.name,
       COALESCE(SUM(CAST(p.price AS DECIMAL(10,2)) * CAST(od.quantity AS INT)), 0) AS Least_Rev_Gen
FROM pizza_types AS pt
INNER JOIN pizzas AS p
ON pt.pizza_type_id = p.pizza_type_id
INNER JOIN order_details AS od
ON od.pizza_id = p.pizza_id
GROUP BY pt.name
ORDER BY Least_Rev_Gen ASC;
```

RESULT!

	name	Least_Rev_Gen
1	The Brie Carre Pizza	23177.00
2	The Green Garden Pizza	27911.50
3	The Spinach Supreme Pizza	30555.50

5. WHAT IS THE AVERAGE ORDER VALUE?

```sql
SELECT
      COALESCE(AVG(Order_Total),0) AS AVG_Order_Value
FROM(
SELECT
       od.order_id,
       COALESCE(SUM(CAST(p.price AS DECIMAL(10,2)) * CAST(od.quantity AS INT)), 0) AS Order_Total

FROM order_details AS od
INNER JOIN pizzas AS p
ON od.pizza_id = p.pizza_id
GROUP BY od.order_id) AS OrderTotals;
```

RESULT!

	AVG_Order_Value
1	76.614524

6. WHICH DAYS OF THE WEEK BRING THE MOST ORDERS?

```sql
SELECT
    DATENAME(WEEKDAY, date) AS Weekday,
    COUNT(*) AS orders
FROM orders
GROUP BY DATENAME(WEEKDAY, date)
ORDER BY orders DESC;
```

RESULT!

	Weekday	orders
1	Friday	3538
2	Thursday	3239
3	Saturday	3158
4	Wednesday	3024
5	Tuesday	2973
6	Monday	2794
7	Sunday	2624

7. WHAT ARE THE PEAK ORDER TIMES DURING THE DAY?

```sql
SELECT
    CAST(
        CASE
            WHEN DATEPART(HOUR, time) = 0 THEN 12
            WHEN DATEPART(HOUR, time) <= 12 THEN DATEPART(HOUR, time)
            ELSE DATEPART(HOUR, time) - 12
        END AS VARCHAR(2)
    )
    + ' ' +
    CASE
        WHEN DATEPART(HOUR, time) < 12 THEN 'AM'
        ELSE 'PM'
    END AS Peak_Hour,

    COUNT(*) AS Orders
FROM orders
GROUP BY DATEPART(HOUR, time)
ORDER BY Orders DESC;
```

RESULT!

	Peak_Hour	Orders
1	12 PM	2520
2	1 PM	2455
3	6 PM	2399
4	5 PM	2336
5	7 PM	2009
6	4 PM	1920
7	8 PM	1642
8	2 PM	1472
9	3 PM	1468
10	11 AM	1231
11	9 PM	1198
12	10 PM	663
13	11 PM	28
14	10 AM	8
15	9 AM	1

8. DO WEEKENDS GENERATE MORE REVENUE THAN WEEKDAYS?

```
SELECT
    CASE
        WHEN DATENAME(WEEKDAY, o.date) IN ('Saturday', 'Sunday') THEN 'Weekend'
        ELSE 'Weekday'
    END AS Day_Type,
    COALESCE(SUM(CAST(p.price AS DECIMAL(10,2)) * CAST(od.quantity AS INT)), 0) AS Revenue
FROM orders o
INNER JOIN order_details od ON o.order_id = od.order_id
INNER JOIN pizzas p ON od.pizza_id = p.pizza_id
GROUP BY
    CASE
        WHEN DATENAME(WEEKDAY, o.date) IN ('Saturday', 'Sunday') THEN 'Weekend'
        ELSE 'Weekday'
    END
ORDER BY Revenue DESC;
```

RESULT!

	Day_Type	Revenue
1	Weekday	1190948.30
2	Weekend	444771.80

9. HOW DOES SALES VARY BY MONTH/SEASON?

```sql
SELECT
    COALESCE(SUM(CAST(p.price AS DECIMAL(10,2)) * CAST(od.quantity AS INT)), 0) AS Sales,
    DATENAME(MONTH, o.date) AS MonthName,
    MONTH(o.date) AS MonthNumber
FROM order_details AS od
INNER JOIN orders AS o ON od.order_id = o.order_id
INNER JOIN pizzas AS p ON p.pizza_id = od.pizza_id
GROUP BY DATENAME(MONTH, o.date), MONTH(o.date)
ORDER BY MonthNumber;
```

RESULT!

	Sales	MonthName	MonthNumber
1	139586.60	January	1
2	130319.20	February	2
3	140794.20	March	3
4	137473.60	April	4
5	142805.50	May	5
6	136460.40	June	6
7	145115.80	July	7
8	136556.50	August	8
9	128360.10	September	9
10	128055.20	October	10
11	140790.70	November	11
12	129402.30	December	12

10. ARE THERE SPECIFIC HOURS WHEN CERTAIN CATEGORIES (E.G., CHICKEN, VEGGIE) SELL MORE?

```sql
SELECT
    pt.category AS Category,
    DATEPART(HOUR, o.time) AS Hour,
    CASE
        WHEN DATEPART(HOUR, o.time) < 12 THEN 'AM'
        ELSE 'PM'
    END AS AM_PM,
    COALESCE(SUM(CAST(p.price AS DECIMAL(10,2)) * CAST(od.quantity AS INT)), 0) AS Sales
FROM orders AS o
INNER JOIN order_details AS od ON o.order_id = od.order_id
INNER JOIN pizzas AS p ON p.pizza_id = od.pizza_id
INNER JOIN pizza_types AS pt ON pt.pizza_type_id = p.pizza_type_id
GROUP BY pt.category, DATEPART(HOUR, o.time),
         CASE WHEN DATEPART(HOUR, o.time) < 12 THEN 'AM' ELSE 'PM' END
ORDER BY Sales DESC;
```

RESULT!

	Category	Hour	AM_PM	Sales
1	Supreme	12	PM	56565.70
2	Classic	12	PM	54989.70
3	Veggie	12	PM	54852.40
4	Supreme	13	PM	53511.30
5	Classic	13	PM	52975.80
6	Chicken	12	PM	52464.00
7	Chicken	13	PM	50948.00
8	Veggie	13	PM	50242.30
9	Supreme	18	PM	45241.60
10	Classic	18	PM	44558.00
11	Supreme	17	PM	43482.20
12	Classic	17	PM	42994.90
13	Chicken	18	PM	42906.00

11. WHICH PIZZA SIZE IS ORDERED THE MOST?

```sql
SELECT
    p.size,
    COUNT(*) AS Orders
FROM order_details AS od
INNER JOIN pizzas AS p ON od.pizza_id = p.pizza_id
GROUP BY p.size
ORDER BY Orders DESC;
```

RESULT!

	size	Orders
1	L	37052
2	M	30770
3	S	28274
4	XL	1088
5	XXL	56

12. WHICH PIZZA SIZE GENERATES THE MOST REVENUE?

```sql
SELECT
        COALESCE(SUM(CAST(p.price AS DECIMAL(10,2)) * CAST(od.quantity AS INT)), 0) AS Sales,
        p.size
FROM order_details AS od
INNER JOIN pizzas AS p
ON od.pizza_id = p.pizza_id
GROUP BY P.size
ORDER BY Sales DESC;
```

RESULT!

	Sales	size
1	750637.40	L
2	498764.50	M
3	356153.00	S
4	28152.00	XL
5	2013.20	XXL

13. ARE "LARGE" PIZZAS MORE POPULAR ON WEEKENDS COMPARED TO WEEKDAYS?

```sql
SELECT
    CASE
        WHEN DATENAME(WEEKDAY, o.date) IN ('Saturday', 'Sunday') THEN 'Weekend'
        ELSE 'Weekday'
    END AS Days_Of_Week,
    p.size,
    COUNT(*) AS OrdersCount
FROM order_details AS od
INNER JOIN pizzas AS p ON od.pizza_id = p.pizza_id
INNER JOIN orders AS o ON o.order_id = od.order_id
WHERE p.size = 'L'
GROUP BY
    CASE
        WHEN DATENAME(WEEKDAY, o.date) IN ('Saturday', 'Sunday') THEN 'Weekend'
        ELSE 'Weekday'
    END,
    p.size
ORDER BY Days_Of_Week, OrdersCount DESC;
```

RESULT!

	Days_Of_Week	size	OrdersCount
1	Weekday	L	27196
2	Weekend	L	9856

14. WHICH PIZZAS ARE OFTEN ORDERED TOGETHER (BASKET ANALYSIS)?

```sql
SELECT TOP 10
    p1.pizza_id AS Pizza1_ID,
    pt1.category AS Pizza1_Category,
    p2.pizza_id AS Pizza2_ID,
    pt2.category AS Pizza2_Category,
    COUNT(*) AS Times_Ordered_Together
FROM order_details od1
INNER JOIN order_details od2
ON od1.order_id = od2.order_id
    AND od1.pizza_id < od2.pizza_id
INNER JOIN pizzas p1
ON od1.pizza_id = p1.pizza_id
INNER JOIN pizza_types pt1
ON p1.pizza_type_id = pt1.pizza_type_id
INNER JOIN pizzas p2
ON od2.pizza_id = p2.pizza_id
INNER JOIN pizza_types pt2
ON p2.pizza_type_id = pt2.pizza_type_id
GROUP BY
    p1.pizza_id, pt1.category,
    p2.pizza_id, pt2.category
ORDER BY Times_Ordered_Together DESC
```

RESULT!

	Pizza1_ID	Pizza1_Category	Pizza2_ID	Pizza2_Category	Times_Ordered_Together
1	big_meat_s	Classic	thai_ckn_l	Chicken	576
2	big_meat_s	Classic	four_cheese_l	Veggie	504
3	big_meat_s	Classic	five_cheese_l	Veggie	500
4	big_meat_s	Classic	classic_dlx_m	Classic	420
5	big_meat_s	Classic	spicy_ital_l	Supreme	416
6	big_meat_s	Classic	cali_ckn_l	Chicken	392
7	five_cheese_l	Veggie	thai_ckn_l	Chicken	392
8	big_meat_s	Classic	southw_ckn_l	Chicken	380
9	big_meat_s	Classic	hawaiian_s	Classic	356
10	five_cheese_l	Veggie	spicy_ital_l	Supreme	356

15. ARE SPICY PIZZAS MORE POPULAR THAN NON-SPICY ONES?

```sql
SELECT
    CASE
        WHEN pt.category LIKE '%Spicy%' OR pt.name LIKE '%Spicy%' THEN 'Spicy'
        ELSE 'Non-Spicy'
    END AS Spicy_Status,
    COUNT(*) AS Number_Of_Orders,
    SUM(CAST(od.quantity AS INT)) AS Total_Quantity
FROM order_details od
INNER JOIN pizzas p ON od.pizza_id = p.pizza_id
INNER JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY
    CASE
        WHEN pt.category LIKE '%Spicy%' OR pt.name LIKE '%Spicy%' THEN 'Spicy'
        ELSE 'Non-Spicy'
    END
ORDER BY Total_Quantity DESC;
```

RESULT!

	Spicy_Status	Number_Of_Orders	Total_Quantity
1	Non-Spicy	93466	95300
2	Spicy	3774	3848

16. WHICH PIZZAS HAVE HIGH QUANTITY SALES BUT LOW REVENUE (CHEAP BUT POPULAR)?

```sql
SELECT
    pt.name AS pizza_name,
    SUM(CAST(od.quantity AS INT)) AS total_quantity_sold,
    SUM(CAST(od.quantity AS INT) * CAST(p.price AS DECIMAL(10,2))) AS total_revenue,
    AVG(CAST(p.price AS DECIMAL(10,2))) AS avg_price
FROM pizzas p
JOIN pizza_types pt
    ON p.pizza_type_id = pt.pizza_type_id
JOIN order_details od
    ON p.pizza_id = od.pizza_id
GROUP BY pt.name
HAVING SUM(CAST(od.quantity AS INT)) > (
        SELECT AVG(total_quantity)
        FROM (
            SELECT SUM(CAST(od2.quantity AS INT)) AS total_quantity
            FROM order_details od2
            JOIN pizzas p2 ON od2.pizza_id = p2.pizza_id
            GROUP BY p2.pizza_type_id
        ) AS sub
    )
  AND SUM(CAST(od.quantity AS INT) * CAST(p.price AS DECIMAL(10,2))) < (
        SELECT AVG(total_revenue)
        FROM (
            SELECT SUM(CAST(od3.quantity AS INT) * CAST(p3.price AS DECIMAL(10,2))) AS total_revenue
            FROM order_details od3
            JOIN pizzas p3 ON od3.pizza_id = p3.pizza_id
            GROUP BY p3.pizza_type_id
        ) AS sub2
    )
ORDER BY total_quantity_sold DESC, total_revenue ASC;
```

RESULT!

	pizza_name	total_quantity_sold	total_revenue	avg_price
1	The Big Meat Pizza	3828	45936.00	12.000000

17. WHICH PIZZAS HAVE LOW QUANTITY SALES BUT HIGH REVENUE (PREMIUM PIZZAS)?

```sql
SELECT
    pt.name AS pizza_name,
    SUM(CAST(od.quantity AS INT)) AS total_quantity_sold,
    SUM(CAST(od.quantity AS INT) * CAST(p.price AS DECIMAL(10,2))) AS total_revenue,
    AVG(CAST(p.price AS DECIMAL(10,2))) AS avg_price
FROM pizzas p
JOIN pizza_types pt
    ON p.pizza_type_id = pt.pizza_type_id
JOIN order_details od
    ON p.pizza_id = od.pizza_id
GROUP BY pt.name
HAVING SUM(CAST(od.quantity AS INT)) < (
           SELECT AVG(total_quantity)
           FROM (
                SELECT SUM(CAST(od2.quantity AS INT)) AS total_quantity
                FROM order_details od2
                JOIN pizzas p2 ON od2.pizza_id = p2.pizza_id
                GROUP BY p2.pizza_type_id
           ) AS sub
       )
   AND SUM(CAST(od.quantity AS INT) * CAST(p.price AS DECIMAL(10,2))) > (
           SELECT AVG(total_revenue)
           FROM (
                SELECT SUM(CAST(od3.quantity AS INT) * CAST(p3.price AS DECIMAL(10,2))) AS total_revenue
                FROM order_details od3
                JOIN pizzas p3 ON od3.pizza_id = p3.pizza_id
                GROUP BY p3.pizza_type_id
           ) AS sub2
       )
ORDER BY total_revenue DESC, total_quantity_sold ASC;
```

RESULT!

	pizza_name	total_quantity_sold	total_revenue	avg_price
1	The Greek Pizza	2840	56908.20	20.035633
2	The Mexicana Pizza	2968	53561.50	18.024038
3	The Five Cheese Pizza	2818	52133.00	18.500000

18. WHAT PERCENTAGE OF TOTAL SALES IS CONTRIBUTED BY THE TOP 3 PIZZAS (PARETO PRINCIPLE)?

```sql
WITH pizza_sales AS (
    SELECT
        pt.name AS pizza_name,
        SUM(CAST(od.quantity AS INT) * CAST(p.price AS DECIMAL(10,2))) AS total_revenue
    FROM pizzas p
    JOIN pizza_types pt
        ON p.pizza_type_id = pt.pizza_type_id
    JOIN order_details od
        ON p.pizza_id = od.pizza_id
    GROUP BY pt.name
),
ranked_sales AS (
    SELECT
        pizza_name,
        total_revenue,
        ROW_NUMBER() OVER (ORDER BY total_revenue DESC) AS rn
    FROM pizza_sales
)
SELECT
    ROUND(
        (SUM(CASE WHEN rn <= 3 THEN total_revenue ELSE 0 END) * 100.0) /
         SUM(total_revenue),
        2
    ) AS top_3_percentage
FROM ranked_sales;
```

RESULT!

	top_3_percentage
1	15.600000

19. WHICH CATEGORIES (CLASSIC, VEGGIE, CHICKEN, SUPREME) ARE UNDERPERFORMING?

```sql
WITH category_sales AS (
    SELECT
        pt.category,
        SUM(CAST(od.quantity AS INT) * CAST(p.price AS DECIMAL(10,2))) AS total_revenue,
        SUM(CAST(od.quantity AS INT)) AS total_quantity
    FROM pizzas p
    JOIN pizza_types pt
        ON p.pizza_type_id = pt.pizza_type_id
    JOIN order_details od
        ON p.pizza_id = od.pizza_id
    GROUP BY pt.category
)
SELECT
    category,
    total_revenue,
    total_quantity
FROM category_sales
WHERE total_revenue < (SELECT AVG(total_revenue) FROM category_sales)
ORDER BY total_revenue ASC;
```

RESULT!

	category	total_revenue	total_quantity
1	Mushroom	37669.00	2718

20. WHAT'S THE DISTRIBUTION OF ORDERS ACROSS DIFFERENT TIMES OF THE DAY (MORNING, AFTERNOON, EVENING, NIGHT)?

```sql
SELECT
    CASE
        WHEN DATEPART(HOUR, time) BETWEEN 6 AND 11 THEN 'Morning'
        WHEN DATEPART(HOUR, time) BETWEEN 12 AND 16 THEN 'Afternoon'
        WHEN DATEPART(HOUR, time) BETWEEN 17 AND 21 THEN 'Evening'
        ELSE 'Night'
    END AS time_of_day,
    COUNT(order_id) AS total_orders,
    ROUND( (COUNT(order_id) * 100.0 / SUM(COUNT(order_id)) OVER()), 2 ) AS percentage_share
FROM orders
GROUP BY
    CASE
        WHEN DATEPART(HOUR, time) BETWEEN 6 AND 11 THEN 'Morning'
        WHEN DATEPART(HOUR, time) BETWEEN 12 AND 16 THEN 'Afternoon'
        WHEN DATEPART(HOUR, time) BETWEEN 17 AND 21 THEN 'Evening'
        ELSE 'Night'
    END
ORDER BY total_orders DESC;
```

RESULT!

	time_of_day	total_orders	percentage_share
1	Afternoon	9835	46.070000000000
2	Evening	9584	44.890000000000
3	Morning	1240	5.810000000000
4	Night	691	3.240000000000

PIZZA SALES REPORT :

- Top Performers: A small number of premium pizzas generate the majority of revenue, even with lower sales volumes.
- High-Volume Pizzas: Classic and affordable pizzas consistently drive higher order quantities, showing strong customer preference for value options.
- Premium Impact: Certain pizzas with below-average sales but above-average revenue highlight the role of pricing in profitability.
- Category Insights: Categories like Classic dominate in quantity, while Supreme and Chicken pizzas perform better in revenue share.
- Time Trends: Evening hours remain peak sales time, suggesting demand aligns with dinner consumption.

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KEY TAKEAWAYS:

Balancing high-volume, affordable pizzas with premium, high-margin options creates a sustainable revenue mix. Strategic pricing and targeted promotions can further optimize sales.

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